PYTHON LAB MANUAL

ASSIGNMENT-1

1. WAP to enter two integer from the keyboard and perform all the arithmetic operations on them.

Ans:

```
x=int(input("enter the value of x:"))
y=int(input("enter the value of y:"))
print("Addition is :", x+y)
print("Subtraction is :",x-y)
print("Multiplication is :",x*y)
print("Division is :",x/y)
print("Reminder is :", x%y)
print("Floor value is :",x//y)
```

Output:

```
>>> %Run 'Q1 Arithimetic.py'
 enter the value of x:15
 enter the value of y:10
 addition is : 25
 substraction is : 5
 Multiplication is : 150
 division is : 1.5
 reminder is : 5
 floor value is : 1
```

2. WAP to enter three sides of a tringle. Find out the area of the tringle.

Ans:

```
import math
a=int(input("enter the value of a:"))
b=int(input("enter the value of b:"))
c=int(input("enter the value of c:"))
s=float((a+b+c)/2)
z=math.sqrt(s*(s-a)*(s-b)*(s-c))
print("area of tringle is :",z)
```

Output:

```
>>> %Run 'Q2 Area of tringle.py'
 enter the value of a:6
 enter the value of b:8
 enter the value of c:10
 area of tringle is : 24.0
```

3. WAP to enter an integer number and find out its binary, octal and hexadecimal equivalents.

Ans:

```
x=int(input("enter a number:"))
print(x)
print("binary equivalent is :",bin(x))
print("octal equivalent is :",oct(x))
print("hexadecimal equivalent is :",hex(x))
```

Output:

```
>>> %Run 'Q3 Conversion.py'
 enter a number:7
 7
 binary equivalent is : 0b111
 octal equivalent is : 0o7
 hexadecimal equivalent is : 0x7
```

4.WAP to enter the coefficients of a quadratic equation and findout the roots of the equivalent

```
import math
a=int(input("enter the value of a:"))
b=int(input("enter the value of b:"))
c=int(input("enter the value of c:"))
d=math.sqrt(b*b)-(4*c*a)
x=(-b+d)/(2*a)
y=(-b-d)/(2*a)
print("the roots are:",x,"&",y)
```

Output:

```
>>> %Run 'Q4 Quadratic.py'
 enter the value of a:6
 enter the value of b:5
 enter the value of c:7
 the roots are: -14.0 & 13.17
```

5. WAP to enter a temperature in farenheite and convert it into celsiuss.

Ans:

```
F=float(input("enter the temp. in farenheite:"))
C= ((F-32)*5)/9
print("The value of farenheite in celsius is :",C)
```

Output:

```
>>> %Run 'Q5 farenthit to celcius.py'
 enter the temp. in farenheite:98.6
 The value of farenheite in celsius is : 37.0
```

6. WAP to convert distance in meters to Kilometers.

Ans:

```
x=int(input("Enter the distance in meter:"))
y=int(x/1000)
z=int(x%1000)
print(y,"kilometer",z,"meter")
```

Output:

```
>>> %Run 'Q6 Distacne.py'
 Enter the distance in meter:2536
 2 kilometer 536 meter
```

7. WAP to enter time in minutes and print it in hour and minute format.

Ans:

```
x=int(input("enter the time:"))
h=int(x/60)
m=x%60
print(h,"hour",m,"minute")
```

Output:

```
>>> %Run 'Q7 Minute.py'
 enter the time in minutes:2436
 40 hour 36 minute
```

8. WAP to enter two numbers nd swap them using bitwise operators.

Ans:

```
a=int(input("enter the value of a:"))
```

```
b=int(input("enter the value of b:"))
a=a^b
b=a^b
a=a^b
print("after convesion the value of a:",a)
print("after convesion the value of b:",b)
```

Output:

```
>>> %Run 'Q8 Swap Numbers.py'
 enter the value of a:15
 enter the value of b:12
 after convesion the value of a: 12
 after convesion the value of b: 15
```

ASSIGNMENT-2

1. WAP to enter a percentage and display the gradation system.

Ans:

```
x=int(input("enter the percentage:"))
if x>=90:
   print("the equivalent grade is 'O'")
elif x>=80 and x<90:
   print("the equivalent grade is 'E'")
elif x>=70 and x<80:
   print("the equivalent grade is 'A'")
elif x>=60 and x<70:
   print("the equivalent grade is 'B'")
elif x>=50 and x<60:
   print("the equivalent grade is 'C'")
elif x>=40 and x<50:
   print("the equivalent grade is 'D'")
else:
   print("the equivalent grade is 'F'")
```

2. WAP to enter a 5 digit number and print the digits present at odd locations.

Ans:

```
I=[]
x=int(input("Enter a 5 digit number:"))
while(x!=0):
  y=x%10
  x=int(x/10)
  I.append(y)
print("Numbers at odd postion :")
for i in range(len(I)-1,-1,-2):
    print(I[i])
```

3. WAP to enter a number and check wheather it is a palindrome number or not.

Ans:

```
x=int(input("enter a number:"))
temp=x
rev=0
while x>0:
    y=x%10
    rev=rev*10+y
    x=x//10
if temp==rev:
    print("It is a Palindrome number")
else:
    print("It is not a Palindorme number")
```

4. **WAP to enter a number and check whether the number is perfect number or not.**

Ans:

```
x=int(input("enter a number:"))
s=0
for i in range(1,x):
    if(x%i==0):
        s=s+i
if(s==x):
```

```
    print("It is a perfect number")
else:
    print("It is not a perfect number")
```

5. WAP to enter a string and print it in reverse also print the no of vowels and consonant in it.

Ans:

```
x=input("enter a string:")
I=x.split()
I1=[]
i=len(I)-1
while i>=0:
    I1.append(I[i])
    i=i-1
output=' '.join(I1)
print(output)
vowels=0
consonant=0
for i in x:
    if(i=='a' or i=='e' or i=='i' or i=='o' or i=='u' or i=='A' or i=='E' or i=='I' or i=='O' or i=='U'):
        vowels=vowels+1
    else:
        consonant=consonant+1
print("The vowels are:",vowels)
print("The consonants are:",consonant)
```

6. WAP to enter a 3 digit number and print all the prime factors of the number.

Ans:

```
x=int(input("Enter a 3 digit numbers:"))
c=2
while(x>1):
    if(x%c == 0):
        print(c,end=" ")
```

```
        x=x/c
    else:
   c=c+1
```

ASSIGNMENT-3

1. WAP to print twin prime numbers between 1 to N.

Ans:

```
N=int(input("Enter a number:"))
for i in range(2,N):
   d=0
   for j in range(1,i+1):
       if i%j==0:
           d=d+1
   if d==2:
       d=0
       N=i+2
       for j in range(1,N+1):
           if N%j==0:
               d=d+1
       if d==2:
           print("(%d,%d)"%(i,N))
```

2. WAP to find the factorial of a number.

Ans:

```
x=int(input("Enter a number:"))
factorial=1
if x<0:
    print("It is a negative number")
```

```
elif x==0:
    print("The factorial of ",x,"is :",factorial)
else:
    for i in range(1,x+1):
      factorial=factorial*i
    print("The factorial of",x,"is:",factorial)
```

3. WAP to check the leap year.

Ans:

```
x=int(input("Enter a year:"))
if(x%4 == 0) and (x%100 != 0) or (x%400==0):
    print(x,"is a leap year")
else:
    print(x,"is not a leap year")
```

4. WAP to check whether the string is symmetrical or palindrome.

Ans:

```
x=input("enter a string:")
z=(str(str(x)[::-1]))
if x == z:
    print("it is a palindrome")
else:
    print("it is not a palindrome")
```

5. WAP to print even length words in a string.

Ans:

```
x=input("Enter a String:")
y= ""
for c in x:
    if c != " ":
        y=y+c
```

```
    else:
        if len(y) % 2 == 0:
            print(y)
        y = ""
if len(y) % 2 == 0:
  print(y)
```

6. WAP to remove all duplicates from a given string.

Ans:

```
x=input("Enter a string:")
result = ""
for char in x:
    if char not in result:
        result=result+char
print(result)
```

ASSIGNMENT-4

1. WAP to print the second largest and second smallest element in a list of 10 integers without using sort function.

Ans:

```
arr=[]
x=int(input("enter the no of elements:"))
for i in range(x):
    m=int(input("Enter the element:"))
    arr.append(m)
for j in range(len(arr)-1):
    for k in range(len(arr)-j-1):
        if arr[k] > arr[k+1]:
            arr[k],arr[k+1]=arr[k+1],arr[k]
print("The sorted array is:")
print(arr)
```

```
second_largest=arr[-2]
second_smallest=arr[1]
print("Second largest element is:",second_largest)
print("Second smallest element is:",second_smallest)
```

2. WAP to create two lists first list containing 5 integers and second list containing 5 strings.print both the lists one element from each list combined at a time.

Ans:

```
list1 = [1, 2, 3, 4, 5]
list2 = ["A", "B", "C","D","E"]
s=[]
for i in range(len(list1)):
    s.append(str(list1[i]))
    s.append(list2[i])
print(s)
```

3. WAP to create an integer list of 20 elements increase the odd valued elements by 5.

Ans:

```
s=[]
x=int(input("enter the no of elements:"))
for i in range(x):
    n=int(input("Enter the element:"))
    s.append(n)
for i in range(x):
    if s[i]%2!=0:
        s[i]=s[i]+5
print(s)
```

4. WAP to create a function that prints the first 15 terms of the fibonacci series without using recursion.

Ans:

```
def fibonacci(n):
    a=1
    b = 1
```

```
    print(a,b,end=" ")
   # print(b)
    for i in range(2, n):
        c = a + b
        print(c,end=" ")
        a = b
        b = c
n = int(input("Enter the number of terms you want to print: "))
print("The Fibonacci series of first",n,"terms are:")
fibonacci(n)
```

5. WAP to create a function that takes list as argument and returns the even values of the list.Print the new list with even values.

Ans:

```
def even_len(n):
    s=[]
    for i in n:
        if i%2==0:
            s.append(i)
    return s
x=input("enter the list elements:")
n=[int(i) for i in x.split(",")]S
s=even_len(n)
print(s)
```

6. WAP to calculate factorial of a number using recursion.

Ans:

```
def factorial(n):
    if n==0:
        result=1
    else:
        result=n*factorial(n-1)
    return result
```

```
n=int(input("Enter the number:"))
if(n<0):
    print("It is a Negative number:")
else:
    print("Factorial of",n,"is:",factorial(n))
```

ASSIGNMENT-5

1. WAP to input two dictionaries and print the values by merging the two dictionaries.

Ans:

```
dict1=dict(eval(input("Enter a dictionary values:")))
print(dict1)
dict2=dict(eval(input("Enter a dictionary values:")))
print(dict2)
dict1.update(dict2)
print("After merging the list is:")
print(dict1)
```

2. WAP to create a dictionary and print the key which has the maximum unique value.

Ans:

```
d = dict(eval(input("Enter the dictionary: ")))
s = []
s1 = []
for v in d.keys():
    s.append(v)
for b in d.values():
    s1.append(b)
mv = s1[0]
mk = s[0]
for i in range(1, len(s)):
    if s1[i] > mv:
```

```
        mv = s1[i]
        mk = s[i]
print("The key has maximum unique value is:",mk)
```

3. WAP to enter a dictionary and remove the duplicate value inside the dictionary.

Ans:

```
d=dict(eval(input("Enter the dictionary:")))
temp = []
res = dict()
for key, val in d.items():
    if val not in temp:
        temp.append(val)
        res[key] = val
print(res)
```

4. WAP to enter a set and copy the content of the set into a new set one element at a time.

Ans:

```
s=set(eval(input("Enter a set element:")))
s1=set()
for i in range(1,len(s)+1):
    s1.update(s)
print("Updated set is :",s1);
```

5. WAP to enter two sets and perform alll the set operations on it.

Ans:

```
s1=set(eval(input('Enter the 1st set:')))
s2=set(eval(input('Enter the 2nd set:')))
print("Union operation is:",s1.union(s2))
print("Intersection operation is:",s1.intersection(s2))
print("The difference is:",s1.difference(s2))
print("The symmetric_difference is:",s1.symmetric_difference(s2))
```

6. WAP to enter two different sets with string elements combine both the sets remove any duplicates are present, print the new set.

Ans:

```
s2=set(eval(input('Enter the 2nd set:')))
s3=set()
s1.update(s2)
s3=s1
print("After updation the new set is:",s3)
```

ASSIGNMENT-6

1. WAP to print a matrix containing group of similar elements by giving the input randomly in another matrix.

Ans:

```
from collections import Counter
test_list=[1,3,5,1,3,2,5,4,2,1]
print("The original lis:"+str(test_list))
temp=Counter(test_list)
res=[[key]*val for key,val in temp.items()]
print("Matrix after grouping:"+str(res))
```

2. WAP to print a matrix along with a sumation of row elements and column elements after entering a 3*3 matrix.

Ans:

```
a = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
b = [[0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, " "]]
rows = len(a)
cols = len(a[0])
for i in range(0, rows):
    sumRow = 0
    for j in range(0, cols):
        sumRow = sumRow + a[i][j]
        b[i][j] = a[i][j]
        b[i][cols] = sumRow
for i in range(0, cols):
```

```
    sumCol = 0
    for j in range(0, rows):
        sumCol = sumCol + a[j][i]
        b[rows][i] = sumCol
for i in range(0,rows+1):
    for j in range(0,cols+1):
        print(b[i][j],"\t",end=" ")
    print()
```

3. WAP to print the transpose of a matrix of n*n order.

Ans:

```
Rows=int(input("Give the number of rows:"))
Columns=int(input("Give the number of columns:"))
matrix=[[int(input()) for c in range(Columns)] for r in range(Rows)]
print(Rows)
print("Original Matrix")
for i in range(0,Rows):
    for j in range(0,Columns):
        print(matrix[i][j],"\t",end=" ")
    print()
print("Matrix after transpose")
for i in range(0,Rows):
    for j in range(0,Columns):
        print(matrix[j][i],"\t",end="")
    print()
```

4. WAP to check if a value is present in the list or not, using lambda function.

Ans:

```
x=list(eval(input("Enter the list:")))
y=int(input("enter the element you want to search:"))
z = lambda x: (x == y)
found = False
```

```
for e in x:
    if z(e):
        found = True
        break
if found:
    print(y, "is present in the list.")
else:
    print(y, "is not present in the list.")
```

5. WAP to print Fibonacci series up to n terms using lambda function.

Ans:

```
n = int(input("Enter the number of terms: "))
Fibonacci = lambda n: [0] if n == 1 else [0, 1] if n == 2 else Fibonacci(n-1) + [Fibonacci(n-1)[-1] + Fibonacci(n-2)[-1]]
print("Fibonacci series:")
for i in range(n):
    print(fibonacci(i+1)[-1], end=" ")
```

6. WAP to print the intersection of 2 arrays using lambda function.

Ans:

```
array1=list(eval(input("Enter the 1st list:")))
array2=list(eval(input("Enter the 2nd list:")))
intersection = list(filter(lambda x: x in array1, array2))
print("Intersection of array1 & array2 is: ", intersection)
```

ASSIGNMENT-7

1. WAP to triple all numbers in a given list of integers. Use map()

Ans:

```
x=list(eval(input("Enter a list:")))
def triple(x):
    return x*3
y=list(map(triple,x))
print(y)
```

2. WAP to add three given list using python map and lambda.

Ans:

```
x=list(eval(input("Enter the 1st list:")))
y=list(eval(input("Enter the 2nd list:")))
z=list(eval(input("Enter the 3rd list:")))
print("Original list:")
print(x)
print(y)
print(z)
l1=list(map(lambda x,y,z:x+y+z,x,y,z))
print("The resulttant list is:",l1)
```

3. WAP to create a list containing the power of said number in bases raised to the corresponding number in the index using python map.

Ans:

```
from math import *
x=list(eval(input("Enter the power list:")))
y=list(eval(input("enter the number list:")))
def power(x,y):
    z=int(pow(y,x))
    return z
l=list(map(power,x,y))
print("The resultant list:",l)
```

4. WAP to convert all the characters into uppercase and lowercase and eliminate duplicate letters from a given sequence. Use map() function.

Ans:

```
def swap(y):
    if (y.isupper()):
        return str(y).lower()
    else:
        return str(y).upper()
y=list(eval(input("Enter the character list:")))
```

```
s=map(swap,y)
print(set(s))
```

5. WAP to convert a given list of integer and a tuple of integer in a list of string using map().

Ans:

```
def convert(c):
    if c==1:
        return "one"
    elif c==2:
        return "Two"
    elif c==3:
        return "three"
    elif c==4:
        return "four"
    elif c==5:
        return "five"
    elif c==6:
        return "six"
    elif  c==7:
        return "Seven"
    elif c==8:
        return "eight"
    elif c==9:
        return "nine"
    else:
        return "zero"
l1=list(eval(input("Enter the number list:")))
l2=list(map(convert,l1))
print(l2)
t1=tuple(eval(input("enter the number tuple:")))
t2=list(map(convert,l2))
```

```
print(l2)
```

6. WAP to find the ratio of positive numbers, negative numbers and zeroes in an array of integers using map().

Ans:

```
x=list(eval(input("Enter the integer array:")))
print("The original list:",x)
pos=list(filter(lambda y:y>0,x))
zero=list(filter(lambda y:y==0,x))
neg=list(filter(lambda y:y<0,x))
n=len(x)
z1=[]
P_ratio=round(len(pos)/n,2)
N_ratio=round(len(neg)/n,2)
Z_ratio=round(len(zero)/n,2)
z1.append(P_ratio)
z1.append(N_ratio)
z1.append(Z_ratio)
def ratio(z1):
    return z1
l1=list(map(ratio,z1))
print("The ratio array:",l1)
```

ASSIGNMENT-8

1. WAP for `ndarray` Object, Indexing, and Slicing.

Ans:

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print("Array:\n", arr)
arr = np.array([[10, 20, 30], [40, 50, 60]])
print("Element at (0, 1):", arr[0, 1])  # Indexing
print("Row 1:", arr[1, :])  # Slicing rows
print("Column 2:", arr[:, 2])  # Slicing columns
```

Ans:

2. WAP for Data Types and Structures in NumPy.

Ans:

```
arr = np.array([1, 2, 3], dtype='int32')
print("Data type:", arr.dtype)
arr = arr.astype('float64')
print("Updated type:", arr.dtype)
```

3. WAP for NumPy Array Properties and Functions.

Ans:

```
ones_array = np.ones((2, 3))
zeros_array = np.zeros((2, 3))
empty_array = np.empty((2, 3))
print("Ones:\n", ones_array)
print("Zeros:\n", zeros_array)
print("Empty:\n", empty_array)
arr = np.array([[1, 2, 3], [4, 5, 6]])
print("Shape of array:", arr.shape)
reshaped = arr.reshape(3, 2)
print("Reshaped array:\n", reshaped)
```

```
arr1 = np.array([1, 2, 3])
copy_arr1 = arr1.copy()
view_arr1 = arr1.view()
print("Original:", arr1)
print("Copy:", copy_arr1)
print("View:", view_arr1)
arr2 = np.array([1, 2])
arr3 = np.array([3, 4])
concatenated = np.concatenate((arr2, arr3))
print("Concatenated array:", concatenated)
arr4 = np.array([5, 2, 9, 1])
print("Sorted array:", np.sort(arr4))
```

4. WAP for Statistical Operations and Broadcasting on Arrays.

Ans:

```
arr = np.array([1, 2, 3, 4])
print("Max:", arr.max())
print("Min:", arr.min())
```

```
print("Sum:", arr.sum())

print("Product:", arr.prod())
print("Broadcasted result:", arr + 5)
```

5. WAP for Saving and Loading Arrays.

Ans:

```
arr = np.array([1, 2, 3, 4])

np.save('my_array', arr)  # Save array

loaded_arr = np.load('my_array.npy')  # Load array

print("Loaded array:", loaded_arr)
```

ASSIGNMENT-9

1. WAP to create and work with a Pandas Series.

Ans:

```
import pandas as pd

# Create a Series

data = [10, 20, 30]

labels = ['a', 'b', 'c']

series = pd.Series(data, index=labels)

print("Pandas Series:")

print(series)

# Access data using label

print("Value at label 'b':", series['b'])
```

2. WAP to create and manipulate a data frame.

```
# Create a DataFrame
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
    'City': ['New York', 'San Francisco', 'Los Angeles']
}
df = pd.DataFrame(data)

print("DataFrame:")
print(df)

# Access a column
print("\nAge column:")
print(df['Age'])

# Access a row
print("\nRow at index 1:")
print(df.iloc[1])
```

3. WAP to to read CSV and JSON files.

Ans:

```
# Reading a CSV file

df_csv = pd.read_csv('sample.csv')  # Replace 'sample.csv' with your file path

print("DataFrame from CSV:")
```

```
print(df_csv)


# Reading a JSON file
df_json = pd.read_json('sample.json')  # Replace 'sample.json' with your file path
print("\nDataFrame from JSON:")
print(df_json)
```

4. WAP to calculate the correlation between columns in a data frame.

Ans:

```
# Create a DataFrame with numerical data
data = {
    'Math': [90, 85, 80, 95],
    'Science': [88, 82, 85, 90],
    'English': [78, 75, 80, 85]
}
df = pd.DataFrame(data)


print("DataFrame:")
print(df)


# Calculate correlation
correlation = df.corr()
print("\nCorrelation Matrix:")
print(correlation)
```

ASSIGNMENT-10

1. WAP to plot a simple graph using `plot()`.

Ans:

```
import matplotlib.pyplot as plt
# Plot a simple line graph
x = [1, 2, 3, 4]
```

```
y = [10, 20, 25, 30]
plt.plot(x, y)
plt.title("Simple Line Plot")
plt.plot(x, y, linestyle='--', color='r', marker='o', label="Data Line")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.legend()
plt.grid()
plt.show()
```

2. WAP to draw the scatter and bar plot.

Ans:

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [5, 7, 6, 8, 7]
plt.scatter(x, y, color='b', label="Scatter Points")
plt.title("Scatter Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.legend()
plt.show()
```

```
import matplotlib.pyplot as plt
categories = ['A', 'B', 'C', 'D']
values = [3, 7, 8, 5]
plt.bar(categories, values, color='g', label="Bar Data")
plt.title("Bar Plot")
plt.xlabel("Categories")
plt.ylabel("Values")
plt.legend()
plt.show()
```

3. WAP to draw Histograms and Box plots.

Ans:

```
import matplotlib.pyplot as plt
data = [22, 87, 5, 43, 56, 73, 55, 54, 11, 20, 51, 5]
plt.hist(data, bins=5, color='purple', label="Histogram Data")
plt.title("Histogram")
```

```
import matplotlib.pyplot as plt
data = [7, 8, 5, 6, 9, 10, 15, 20, 21]
plt.boxplot(data)
plt.title("Box Plot")
plt.show()
```

```
plt.xlabel("Bins")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

4. WAP to draw Pie Charts and Contour plots.

Ans:

```
import matplotlib.pyplot as plt
sizes = [15, 30, 45, 10]
labels = ['A', 'B', 'C', 'D']
colors = ['gold', 'yellowgreen', 'lightcoral', 'lightskyblue']
explode = (0, 0.1, 0, 0)  # Explode 2nd slice
plt.pie(sizes, explode=explode, labels=labels, colors=colors, autopct='%1.1f%%',
shadow=True, startangle=140)
plt.title("Pie Chart")
plt.axis('equal')
plt.show()
```

```
import matplotlib.pyplot as plt
import numpy as np
x = np.linspace(-5, 5, 50)
y = np.linspace(-5, 5, 50)
X, Y = np.meshgrid(x, y)
Z = np.sin(np.sqrt(X**2 + Y**2))
plt.contour(X, Y, Z, levels=10, cmap='viridis')
plt.title("Contour Plot")
plt.colorbar()
plt.show()
```

5. WAP to draw the 3D-plot.
Ans:

```
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure()

ax = fig.add_subplot(111, projection='3d')

x = [1, 2, 3, 4, 5]

y = [5, 6, 7, 8, 9]

z = [2, 3, 3, 3, 2]

ax.scatter(x, y, z, color='r', label="3D Points")

ax.set_title("3D Scatter Plot")

ax.set_xlabel("X-axis")

ax.set_ylabel("Y-axis")

ax.set_zlabel("Z-axis")

plt.legend()

plt.show()
```